

Topo-DiT: A Cognitive Video Generation Architecture Based on Stratified Routing and Topological Dynamics

Abstract Current video generation architectures based on Diffusion models and Transformers (DiT) face a sequence length explosion related to resolution and frame rate, commonly referred to as the "long-sequence memory wall". Furthermore, traditional pixel-level denoising methods lack an intrinsic understanding of the physical movement laws of real-world entities. To address these issues, this paper proposes Topo-DiT, a novel cognitive video generation architecture. This system replicates the "Global Workspace Theory" from cognitive science within the video generation domain. By introducing a stratified cognitive router, it differentially compresses dense spatio-temporal grids into sparse physical topological entities. Topo-DiT consists of four major engines: spatio-temporal perception, cognitive routing, topological dynamics deduction, and holographic broadcast rendering. Physical dynamics prediction is accomplished within a low-dimensional entity space based on Flow Matching. This architecture not only significantly reduces cross-node communication overhead and computational requirements but also endows the model with the capability for object-based intuitive physical reasoning.

Keywords: Video Generation, Cognitive Architecture, Stratified Routing, Topological Dynamics, Flow Matching, Optimal Transport

1. Introduction

The human visual cognitive system does not independently process every photoreceptive pixel on the retina with equal computational power when handling dynamic scenes. Instead, it abstracts visual signals into "entity objects" with topological relationships through an attention bottleneck, subsequently performing physical law predictions within a conscious workspace. However, mainstream video generation foundation models (such as Sora-like DiT architectures) predominantly rely on brute-force Spatio-Temporal Grid partitioning, dividing videos into tens of thousands of Patch Tokens for global attention computation. This approach not only severely restricts computational and communication bandwidth but also makes models prone to generating physical hallucinations—such as object melting or unnatural deformations—during long video generation.

The Topo-DiT architecture proposed in this paper aims to transform "compute-intensive pixel denoising" into "logic-intensive element dynamics deduction". By forcibly forming objectified and topological feature clusters in the intermediate layers of the network, this architecture establishes an interface for physical common sense and symbolic logic within large models.

2. Overall System Architecture

The entire Topo-DiT system is divided into four major engines from bottom to top, strictly corresponding to the cognitive psychology processes of "perception → conceptualization → logical deduction → embodiment":

- **Spatio-Temporal Perception Engine:** Responsible for compressing continuous pixel fields into high-dimensional physical grids (3D VAE).

- **Stratified Cognitive Router:** The core innovation layer, responsible for mapping dense grids to sparse topological domain entities (Sinkhorn Optimal Transport).
- **Topological Dynamics Engine:** The main computational backbone, which deduces physical and spatio-temporal evolution in a low-dimensional entity domain based on Flow Matching or Diffusion (Latent Element DiT).
- **Holographic Broadcast Renderer:** Relays high-level evolutionary results downwards to reconstruct high-frequency details of the physical grid (Residual Broadcast).

3. Core Module Engineering Design

3.1 Spatio-Temporal Perception Engine (Front-end Infrastructure)

This engine receives the raw video stream $V \in \mathbb{R}^{B \times 3 \times T \times H \times W}$ as input. The system utilizes a Causal 3D-VAE (such as MagViT-2 variants), applying 2D convolution for the first frame and causal 3D convolution for subsequent frames to prevent future information leakage. Its output stream consists of continuous latent space grids (Grid Tokens) $X_{grid} \in \mathbb{R}^{B \times N \times D}$, where $N = t \times h \times w$. To guarantee absolute position awareness for the subsequent routing stage, 3D RoPE (Rotary Position Encoding) must be injected here. For exceptionally long videos, the system slices the input into multiple Chunks for streaming.

3.2 Stratified Cognitive Router (Core Bridge)

Achieving physically meaningful "element extraction" without breaking end-to-end differentiability remains the core engineering challenge of this module. A standard Softmax operation would lead to "mode collapse" (Attention Collapse), where all anchors attach to the brightest object in the frame. Therefore, this module introduces Optimal Transport theory, employing the Sinkhorn-Knopp algorithm to compute the allocation matrix A :

1. Establish K learnable Topological Domain Anchors $E_{init} \in \mathbb{R}^{B \times K \times D}$.
2. Treat the grid X_{grid} as the "supplier" and the anchors E_{init} as the "demander".
3. Compute a Doubly Stochastic Matrix $A \in \mathbb{R}^{N \times K}$ through a few Sinkhorn iterations. This mathematical constraint strictly guarantees that every grid point must be allocated, and every anchor must absorb a certain amount of information, perfectly resolving mode collapse.

The feature Stratification (collapse) formula for this layer is defined as:

$$X_{element} = A^T X_{grid}$$

3.3 Topological Dynamics Engine (Computational Backbone)

Dynamics deduction operates within the low-dimensional space of $X_{element}$ via denoising or Flow Matching, at which point the space contains only K Tokens (typically $K \in [64, 512]$).

- **Conditioning:** Uses AdaLN-Zero to inject the time step t , T5-XXL embeddings of text prompts, and camera trajectories (Camera Pose/Plucker Coordinates) into the K Element Tokens.

- **Deep DiT (Deep Transformer):** Because K is extremely small, this section can be made exceedingly deep (e.g., 48 or even 96 layers). The update formula is:

$$X_{element}^{(t+1)} = \text{DiT-Blocks}(X_{element}^{(t)}, c)$$

- **Physical Prior Injection:** Because operations are conducted at the "element" level, engineering implementations can incorporate hard physical constraint Masks into this layer's Attention matrix (e.g., forcing "background anchors" not to influence the spatial displacement of "foreground entity anchors," but only generate lighting effects).

3.4 Holographic Broadcast Renderer (Residual Reconstruction)

- **Reverse Routing:** Utilizes the identical matrix A generated by the router to broadcast the updated entity features back to the grid:

$$\Delta X_{grid} = AX_{element}^{(t+1)}$$

- **Gated Residual Connection:** Introduces a learnable gated scalar γ to preserve the high-frequency textures extracted by the VAE:

$$X_{out} = X_{grid} + \gamma \cdot \Delta X_{grid}$$

4. Hardware Mapping and Distributed Parallelism Architecture

The Topo-DiT architecture completely shatters the "long-sequence memory wall" faced by current Sora-like models. Current DiT architectures require global All-to-All communication, making network bandwidth a primary bottleneck. By compressing the massive spatial grid into compact entities, Topo-DiT reduces the communication volume between core computational nodes by a factor of $O(N/K)$, making it inherently suitable for ultra-large-scale cluster training across data centers. For GPU clusters (such as the H100), the system adopts a distributed strategy featuring an asymmetric communication-to-computation ratio:

- **3D VAE Encoding:** The sequence length is massive ($T \times H \times W$), characterized by pure convolution with no global dependencies. The recommended strategy is Spatial-Temporal Parallelism (STP), which distributes video chunks to different GPUs and utilizes Halo Exchange to swap edge pixels.
 - **Cognitive Router:** The sequence length is moderate ($N \times K$), characterized by dense matrix multiplication (Sinkhorn). The recommended strategy is Context Parallelism (CP) or Ring-Attention, reducing Grid Tokens distributed across multiple cards to compute the affinity matrix.
 - **Topological Dynamics (DiT):** The sequence length is extremely small (e.g., $K = 256$), characterized by a deep Transformer with high memory access. The recommended strategy is Tensor Parallelism (TP) combined with Pipeline Parallelism (PP). Because sequence K is too short, Sequence Parallelism is unnecessary. After all GPUs collapse the Grid into Elements, they can be Gathered onto a single machine to perform the 48-layer DiT deduction centrally, massively reducing cross-node communication overhead.
 - **Broadcast and Decoding:** The sequence length is moderate ($N \times K$), involving matrix multiplication and deconvolution. The recommended strategy is to Scatter back to individual GPUs for independent decoding using the CP strategy.
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5. Training Objectives and Alignment Strategy

Pure Mean Squared Error (MSE) cannot supervise a complex network containing intermediate routing structures; thus, a multi-objective joint optimization mechanism is employed in engineering.

5.1 Backbone Dynamics: Optimal Transport Flow Matching (OT-Flow Matching)

Compared to DDPM diffusion models, Flow Matching (such as Rectified Flow) is more appropriate for learning manifold evolutions that possess clear topological trajectories. The objective is to predict the velocity field vector v_θ within the Latent space:

$$\mathcal{L}_{FM} = \mathbb{E}_{t, x_0, x_1} [||v_\theta(x_t, t, c) - (x_1 - x_0)||_2^2]$$

5.2 Routing Layer Constraints: Information Entropy and Orthogonality

The generation of matrix A must be forcibly intervened in engineering to ensure its validity:

- **Sparsity Loss (Reward):** Computes the entropy of A along the row direction, penalizing confused allocations ("one grid belongs to all elements") and rewarding discrete allocations ("one grid belongs to exactly one element").
- **Orthogonality:** Penalizes the feature cosine similarity among the K anchors, forcing them to individually govern different topological domains (such as skin tone, clothing, or background lighting).

5.3 Curriculum Learning Pipeline

Because the architecture is extremely difficult to cold-start, direct end-to-end training will immediately cause gradient explosion. The engineering pipeline is divided into the following three stages:

1. **Phase 1 (Freeze DiT, train router only):** Remove time-step noise and input the noise-free video x_0 . Force an auto-encoding reconstruction of **Grid** -> **Element** -> **Grid**. Using only L2 loss, allow the router to learn to compress the video into K elements and perfectly restore it.
2. **Phase 2 (Freeze router, train DiT):** Load the routing weights from Phase 1 and freeze them. Add noise at the Element layer and train Flow Matching, allowing the model to focus on learning object movement physics.
3. **Phase 3 (Full-parameter joint fine-tuning):** Introduce text conditions, unfreeze all weights, and perform joint fine-tuning, prioritizing the optimization of temporal consistency.

6. Discussion and Conclusion

Looking back at theory from an engineering implementation perspective, the Topo-DiT architecture effectively replicates the "Global Workspace Theory" from cognitive science within the video generation domain. The foundational 10,000 Patches mirror the massive subconscious neural signals of the human retina; the intermediate Sinkhorn routing acts as an attention bottleneck; and the extracted K Element Tokens represent the topological concepts that have entered the "global workspace (conscious mind)".

When the model conducts dynamics deduction within the K space, it is fundamentally performing "Object-based Intuitive Physics" reasoning. This engineering architecture successfully converts "compute-intensive

pixel denoising" into "logic-intensive element dynamics deduction". It not only resolves the memory bottleneck associated with long video generation but also establishes a native interface layer for subsequently injecting knowledge graphs and symbolic logic into video generation.